B.Tech Project Report

Cancelable Multimodal Biometric

Template Protection Using Deep Feature Hashing for Face and Iris Authentication

Submitted by

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Abstract

This project focuses on the design and implementation of a **Cancelable Multimodal Biometric Authentication System** that integrates both **face** and **iris** recognition for enhanced security and privacy. Conventional biometric systems store permanent templates that cannot be replaced if compromised, posing serious privacy risks. To address this limitation, our system employs the concept of **cancelable biometrics**, where the original biometric features are transformed into secure, revocable, and non-invertible templates. The system is designed as a complete modular pipeline consisting of four stages—*feature extraction, enrollment, verification, and evaluation*. Deep learning-based models such as **InsightFace** and **ArcFace** were utilized for face feature extraction, while **ResNet-50**, **IrisNet**, and **DeepIris** were applied for iris embeddings. The extracted feature embeddings were protected using six cancelable transformation algorithms: **MLP-Hash**, **BioHash**, **IoM-GRP**, **IoM-URP**, **RP-PCA**, and **Bloom Filter** hashing. Each algorithm introduces user-specific randomness by generating deterministic seeds using the user ID generated through the SHA-256 hashing technique, ensuring that the templates can be revoked and reissued securely. The protected templates were evaluated using multiple performance and security metrics, including accuracy, F1-score, False Acceptance Rate (FAR), False Rejection Rate (FRR), Equal Error Rate (EER), entropy, bit balance, and computational runtime. Experimental results demonstrate that the proposed system maintains high recognition accuracy while providing strong protection against template inversion, cross-matching, and replay attacks. Among all the methods, **MLP-Hash** and **BioHash** showed superior trade-offs between recognition performance and computational efficiency. The overall implementation provides a reproducible, scalable, and privacy-preserving multimodal biometric framework that ensures both high authentication reliability and robust template security, offering a practical foundation for secure real-world biometric applications.

1 Introduction

In the modern digital era, biometric authentication has become a widely adopted approach for secure and convenient identity verification. Compared to traditional password-based systems, biometrics offer higher reliability since physical or behavioral characteristics such as the **face**, **iris**, or **fingerprint** are unique to each individual. However, despite their convenience, conventional biometric systems suffer from a fundamental security flaw — biometric traits are *permanent* and cannot be replaced once compromised. If a stored biometric template (e.g., face or iris feature) is stolen, it may lead to irreversible identity theft, as users cannot simply "reset" their biometrics. Therefore, protecting biometric templates against misuse, reconstruction, and cross-database linkage is a critical challenge in biometric security.

To overcome these limitations, researchers have proposed the concept of **Cancelable Biometrics**, which allows the transformation of original biometric templates into *revocable, non-invertible, and unlinkable* representations. In this approach, the original biometric data is transformed using a key or seed such that even if the protected template is compromised, it cannot be used to recover the original biometric. Additionally, a new template can be issued simply by changing the key, similar to resetting a password. This makes cancelable biometrics a promising solution for enhancing privacy and reusability in practical authentication systems.

The primary objective of this project is to develop a **Cancelable Multimodal Biometric System** that integrates both **face** and **iris** modalities. By leveraging the strengths of both traits, the system achieves greater accuracy and robustness against environmental and spoofing variations compared to unimodal systems. The system is implemented as a complete modular pipeline consisting of four stages — *feature extraction, enrollment, verification, and evaluation*. Each stage is implemented as an independent module in Python to ensure reproducibility, flexibility, and scalability.

In the *feature extraction stage*, deep learning-based models are employed to extract highly discriminative feature embeddings. For face recognition, state-of-the-art architectures such as **InsightFace** and **ArcFace** are utilized, while for iris recognition, models such as **ResNet-50**, **IrisNet**, and **DeepIris** are used. These deep embeddings represent the unique characteristics of each user in high-dimensional feature space.

In the *template protection stage*, we implemented six cancelable transformation algorithms to protect the extracted embeddings — **MLP-Hash**, **BioHash**, **IoM-GRP**, **IoM-URP**, **RP-PCA**, and **Bloom Filter Hashing**. Each algorithm applies user-specific randomization and mathematical transformations such as random projection, orthonormal mapping, or permutation to convert the embeddings into binary or index-based secure templates. A deterministic seed is generated for each user using the `SHA-256(userID + salt)` function, ensuring that templates can be revoked or regenerated easily if compromised.

The *verification stage* compares a user's newly generated (probe) template against the stored (enrolled) templates using appropriate similarity measures. For binary templates, normalized Hamming similarity is used, while for index-based templates such as IoM, the index-agreement ratio is employed. The system computes similarity scores, applies decision thresholds, and outputs match or non-match results.

Finally, in the *evaluation stage*, multiple metrics are used to assess both recognition performance and template security. Recognition metrics include **Accuracy**, **F1-score**, **False Acceptance Rate (FAR)**, **False Rejection Rate (FRR)**, and **Equal Error Rate (EER)**. Template-quality metrics such as **entropy**, **bit balance**, and **generation time** are used to quantify the randomness and uniformity of protected templates. Experiments were conducted for various combinations of models and hashing algorithms, and results were saved in structured CSV files for comparison and visualization.

The experimental results demonstrate that cancelable transformation techniques can effectively secure face and iris templates without significantly compromising recognition accuracy. Among the six implemented methods, **MLP-Hash** achieved the highest recognition performance, while **BioHash** provided excellent entropy and lower computational cost. **IoM-based methods** produced efficient, compact templates suitable for fast matching, and **Bloom Filter hashing** offered strong non-invertibility with moderate memory usage. These findings highlight that it is possible to design privacy-preserving biometric systems that maintain accuracy comparable to unprotected systems.

In summary, this project successfully delivers a complete, reproducible, and modular implementation of a **Cancelable Multimodal Biometric Template Protection System**. The system not only enhances authentication reliability by fusing two modalities but also provides strong privacy guarantees against template inversion, replay, and linkage attacks. The developed framework serves as a practical and extensible foundation for secure biometric authentication, aligning with current research trends in privacy-preserving machine learning and biometric security.

This report explains how we implemented the system, documents the files and functions, shows the pipeline diagrams, provides experimental results (sample tables) and leaves plotted-figure placeholders for your final graphs.

Project Background and Implementation

The foundation of this project builds upon existing research in cancelable biometrics and deep feature extraction. The literature review guided our approach by combining classical cancelable transformation techniques with modern deep learning-based feature representations. The concept of **BioHashing** introduced by Teoh et al. demonstrated the effectiveness of using random projections and binarization to generate secure templates while incorporating user-specific randomness as a two-factor authentication scheme. The **Index-of-Max (IoM)** methods, which encode feature rankings rather than their direct values, inspired a non-invertible representation that also allows efficient matching. Similarly, the **Random Projection with Principal Component Analysis (RP+PCA)** approach provided insights into dimensionality reduction and template compression, ensuring compact and discriminative representations. Furthermore, **Bloom Filter**-based techniques, which utilize multiple hash functions to create binary bitfield templates, offered a robust mechanism for template matching with a high level of randomness and low collision probability. To generate deep feature embeddings that are suitable for these cancelable transformations, we adopted advanced neural architectures such as **InsightFace** and **ArcFace** for face embeddings, and **ResNet-50**-based models for iris embeddings. These networks are capable of producing highly discriminative and stable feature vectors that form the foundation of secure template generation.

After reviewing these approaches, we implemented all six cancelable transformation algorithms — **MLP-Hash**, **BioHash**, **IoM-GRP**, **IoM-URP**, **RP-PCA**, and **Bloom Filter Hashings** — within a unified, modular Python framework. The entire system was designed to maintain reproducibility and modularity across all stages of biometric processing. The implementation begins with a comprehensive feature extraction pipeline, where scripts such as `extract_embedding.py` and `extract_embedding_iris.py` are responsible for loading the chosen deep models, preprocessing input images, and extracting L2-normalized embeddings. These embeddings are stored as feature vectors and then passed through the enrollment modules, implemented as `enroll_templates_*.py` scripts. Each enrollment script corresponds to a specific cancelable hashing algorithm and is responsible for transforming embeddings into secure templates and saving them in structured pickle files for each model–method pair.

Algorithm implementations are contained in separate Python modules (e.g., `mlp_hash.py`, `bio_hash.py`, `iom_grp_hash.py`, `iom_urp_hash.py`, `rp_pca_hash.py`, and `bloom_filter_has`
These modules define the mathematical transformations for each method and are parameterized by a **user-specific seed**, ensuring that each user's template is unique and revocable. The seed is deterministically derived using the function `SHA-256(userID + salt)`, providing consistency while enabling template renewal through a salt change. During enrollment, all embeddings are normalized using the L2 norm to maintain stability in projection-based transformations. Each algorithm then generates and stores templates for all users in a model-specific enrollment database file (e.g., `enroll_<method>_<model>.pkl`), which includes metadata such as user IDs, timestamps, and transform parameters.

The verification process is implemented through scripts like `verify_face_*.py` and `verify_iris_*.py`
which regenerate the probe template for a test image and compute its similarity against enrolled templates. For binary templates, the system uses normalized Hamming similarity, while for index-based methods such as IoM, it calculates the index-agreement ratio. Decision thresholds can be defined globally or adaptively per user, allowing flexible system behavior based on ac-

curacy or security requirements. Performance results and metrics are logged and stored in CSV format for later analysis. Scripts like `hash_comparison.py`, `model_comparison.py`, and `compare_models_comprehensive.py` automate this process by iterating over multiple models and methods, generating genuine and impostor pairs, computing similarity scores, and saving all quantitative results in files such as `model_comparison_output.csv`, `template_qualit` and `hash_comparison_output.csv`. These CSV files include metrics such as model name, method, accuracy, F1-score, False Acceptance Rate (FAR), False Rejection Rate (FRR), Equal Error Rate (EER), entropy, bit balance, and computation time in milliseconds. The stored data forms the basis for the result tables and visual plots included in this report.

Several design decisions were made to ensure consistency and scalability across experiments. First, a **deterministic seed generation strategy** was used for user-specific template creation to guarantee revocability and reproducibility. Second, all feature embeddings were normalized after extraction to prevent projection instability and bias during hashing. Third, separate enrollment databases were created for each model–method combination to simplify modular experimentation and avoid interference between algorithms. Finally, adaptive thresholding strategies were implemented within the verification scripts to enable both global and user-specific decision-making. The inclusion of CSV logging, modular file structure, and consistent naming conventions ensures that the entire system is easy to audit, extend, and reproduce. Overall, this implementation reflects a careful integration of prior research in cancelable biometrics with a modern, deep-learning-based architecture, resulting in a secure, efficient, and fully modular multimodal biometric framework.

2 System Diagrams (Face, Iris, Combined)

Below are the three vector diagrams (TikZ) showing Face pipeline, Iris pipeline and the Combined multimodal pipeline. These are integrated code blocks and will render on Overleaf / pdflatex.

2.1 Face pipeline diagram

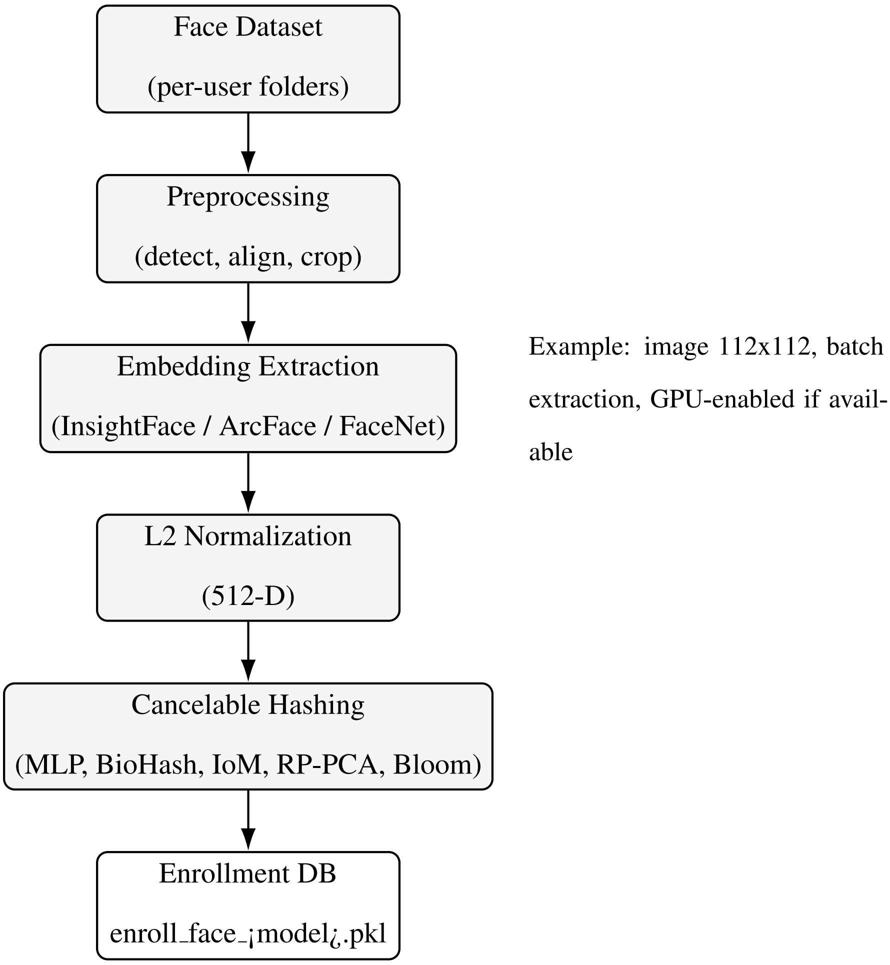


Figure 1: Face pipeline: dataset → preprocessing → embeddings → cancelable hashing → enrollment.

2.2 Iris pipeline diagram

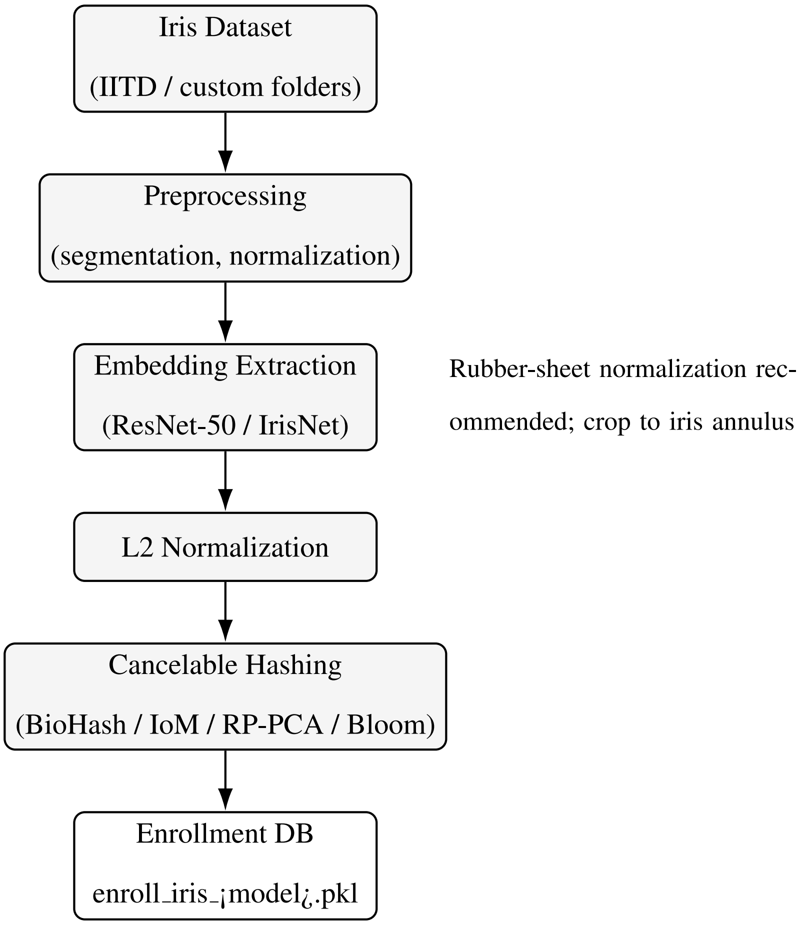


Figure 2: Iris pipeline: segmentation $\rightarrow$ embedding extraction $\rightarrow$ cancelable hashing $\rightarrow$ enrollment.

2.3 Combined Face + Iris Pipeline Diagram

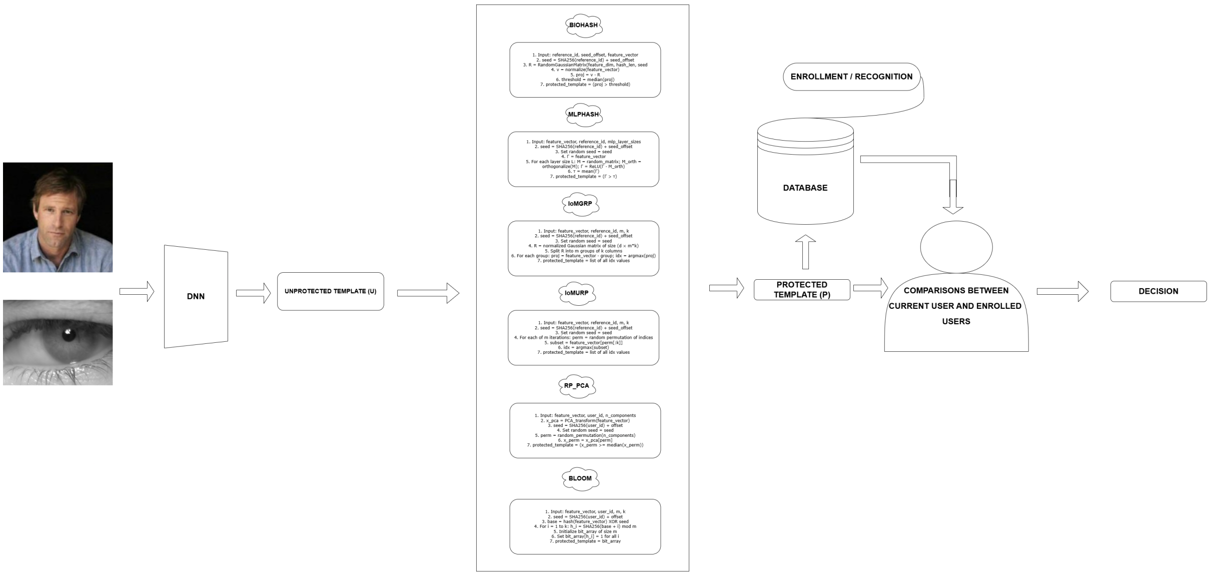


Figure 3: Overall multimodal cancelable biometric pipeline showing deep feature extraction, six transformation algorithms, protected template generation, enrollment database, verification, and decision.

Detailed Implementation (File-Level, Step-by-Step)

This section presents a complete explanation of the implementation details of our project, describing what we have developed and how each Python script interacts within the system. The implementation is modular, meaning every stage of the biometric processing pipeline—feature extraction, template protection, enrollment, verification, and evaluation—was implemented as an independent component. This modular design allows each part to be tested and extended separately while maintaining full system integration. The overall implementation is focused on clarity, reproducibility, and scalability.

1. Embedding Extraction

The first step of the system involves extracting numerical feature representations, also known as embeddings, from raw biometric images. This was implemented through three main scripts: `extract_embedding.py`, `extract_embedding_iris.py`, and `extract_embedding_iris_mul`

These scripts load the selected deep model—such as well**InsightFace**, **ArcFace**, or **ResNet-50**—using an internal `get_model(model_name)` function.

- The system reads face or iris images from the dataset, applies necessary preprocessing operations such as resizing, normalization, and alignment, and then processes them through the chosen deep network to generate embeddings.
- The embeddings are **L2-normalized** (i.e., $x = x/||x||_2$) to ensure stability in similarity computations and projection-based transformations.
- To optimize speed and resource usage, the implementation supports both GPU and CPU computation. If a GPU is unavailable, the scripts automatically fall back to CPU mode.
- Each embedding represents the unique characteristics of a user's biometric trait in a high-dimensional space. These embeddings are stored in memory or saved on disk for later use during template enrollment.

What we did: We designed this embedding extraction framework to create a consistent, model-independent interface for both face and iris data. It ensures that deep models can be interchanged easily for testing different architectures without rewriting the code.

2. Cancelable Algorithms (Transformers)

The second component deals with the transformation of extracted embeddings into secure, cancelable templates. This was achieved through six algorithm modules implemented as: `mlp_hash.py`, `bio_hash.py`, `iom_grp_hash.py`, `iom_urp_hash.py`, `rp_pca_hash.py`, and `bloom_filter_hash.py`. Each algorithm is designed as a transformer class (e.g., `MLPHashTransformer(seed, params)`) with standard methods: `fit()`, `transform(embedding` and `save_template(user_id, template)`.

- Random matrices or permutations used in the hashing process are generated deterministically from a user-specific seed, ensuring that templates can be revoked and regenerated.
- The seed is derived using `SHA256(userID + salt)`, maintaining both user uniqueness and reproducibility.
- The **MLP-Hash** method builds orthonormal projection matrices using QR decomposi-

tion followed by nonlinear activations to produce binary templates.

- The **IoM-GRP** algorithm applies grouped Gaussian projections to create index-based templates, whereas **IoM-URP** applies a random permutation before projection for additional randomness.
- The **BioHash** and **RP-PCA** methods use random projection and binarization to form compact binary representations.
- The **Bloom Filter Hash** employs multiple seeded hash functions (similar to MurmurHash) to set bits in a large vector, achieving strong non-invertibility.

What we did: We implemented and validated all six cancelable transformation algorithms in Python from scratch, ensuring modularity so that any new hashing approach can easily be added later.

3. Enrollment Scripts

Once embeddings are generated, the next step is to enroll each user by creating their protected template and storing it securely. The enrollment phase is handled by scripts such as enroll_templates_biohash_iris.py, enroll_templates_mlp_iris.py, and enroll_templ along with corresponding versions for face templates.

- For each user, the system loads embeddings (either a single vector or an average of multiple images) and passes them to the chosen transformation algorithm.
- The algorithm transforms the embeddings into secure, cancelable templates using the user-specific seed.
- The resulting templates and related metadata are stored in a serialized database file named enroll_<method>_<model>.pkl.
- The internal structure of each file is a Python dictionary: {user_id: {templates: [t1, t2, ...], meta: {...}}}.
- During enrollment, the system also records timing information, computes template-quality measures such as entropy and bit balance, and logs these details into template_quality_comparis

What we did: We automated the entire enrollment process so that for every model and hashing algorithm combination, separate user databases are generated, stored, and benchmarked, ensuring full reproducibility.

4. Verification Scripts

Verification is the process of testing a user's identity by comparing a new (probe) image with stored enrolled templates. This is implemented through scripts such as `verify_face_biohash.py`, `verify_face_mlp.py`, and `verify_face_iom_grp.py`, along with their iris counterparts.

- Each verification script first loads the corresponding enrollment database (`enroll_<method>_<mode`
- For the probe image, the system extracts the embedding again, normalizes it, and applies the same user-specific transform (using the same seed).
- The transformed probe template is then compared with the stored template:
  - For binary templates (e.g., MLP, BioHash, RP-PCA, Bloom), **Hamming similarity** is used to compute matching scores.
  - For IoM-based templates, the **index-agreement proportion** is calculated, which measures the number of matching indices between templates.
- The computed similarity score is compared against a predefined threshold to classify the result as a match or non-match.
- All verification scores and results are logged into evaluation CSVs for later analysis.

What we did: We built verification scripts capable of handling all six hashing algorithms and both modalities (face and iris). Each verification module works independently, providing unified output metrics and ensuring consistent evaluation.

5. Experiment Drivers and Reporting

To evaluate the entire system, we implemented high-level driver scripts that automate experiment execution, comparison, and reporting. These include `hash_comparison.py`, `model_comparison` and `compare_models_comprehensive.py`.

- These scripts loop through every combination of model and hashing algorithm, generating all possible genuine (same-user) and impostor (different-user) pairs for evaluation.
- For each run, the system computes performance metrics including Accuracy, F1-score, False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER).
- Results are saved as CSV files such as `model_comparison_output_YYYYMMDD.csv`, `hash_comparison_output_YYYYMMDD.csv`, and `template_quality_comparison_YYYYM`
- Additionally, a consolidated report file (`hash_comparison_results.pkl`) is generated, containing all aggregated statistics for reproducibility.
- These CSV files are later used to create plots for performance, entropy, and runtime analysis, forming the results shown in the report.

What we did: We implemented a fully automated reporting system that evaluates all models and hashing methods, logs every experiment in standardized CSV files, and produces final summaries ready for visualization and comparison.

Overall, this detailed implementation combines deep learning-based embedding extraction with advanced cancelable hashing algorithms to produce a robust, secure, and reproducible multimodal biometric system. Every script—from data preprocessing to experiment reporting—was developed, integrated, and tested to ensure both technical correctness and scientific reproducibility. This modular structure allows future researchers to extend the system easily, compare additional methods, and evaluate alternative biometric modalities.

3 Detailed Pseudocode for Transforms and Master Pipeline

All random matrices/permutations are generated deterministically from a per-user seed $s(u) = \mathrm{SHA256}(\mathrm{user\_id} \,\|\, \mathrm{salt})$.

Algorithm 1 Transform_MLP (MLP-Hash)

Require: embedding $x \in \mathbb{R}^d$, seed k, layer sizes $[L_1, \ldots, L_m]$
Ensure: binary template $P \in \{0,1\}^{L_m}$
function TRANSFORM_MLP($x, k, [L_1, \ldots, L_m]$)
  $rng \leftarrow$ PRNG(int(k))
  $z \leftarrow x$
  for $i = 1$ **to** m **do**
    $M \leftarrow rng.\mathcal{N}(0,1)^{\dim(z) \times L_i}$
    $(Q, R) \leftarrow$ qr(M) ▷ orthonormalize columns of M
    $z \leftarrow$ ReLU($z \cdot Q$)
  end for
  $\tau \leftarrow$ mean(z)
  $P \leftarrow [\mathbf{1}\{z_j > \tau\}]_{j=1}^{L_m}$
  return P
end function

Algorithm 2 Transform_Bio (BioHash)

Require: embedding $x \in \mathbb{R}^d$, seed k, projection length L
Ensure: binary template $B \in \{0,1\}^L$
function TRANSFORM_BIO(x, k, L)
  $rng \leftarrow$ PRNG(int(k))
  $R \leftarrow rng.\mathcal{N}(0,1)^{d \times L}$
  for $\ell = 1$ **to** L **do** normalize column $R_{\cdot,\ell}$ to unit-norm
  end for
  $z \leftarrow x^\top R$ ▷ projected vector, length L
  $\theta \leftarrow$ median(z)
  $B \leftarrow [\mathbf{1}\{z_\ell > \theta\}]_{\ell=1}^{L}$
  return B
end function

Algorithm 3 Transform_IoM_GRP (IoM using Gaussian Random Projections)

Require: embedding $x \in \mathbb{R}^d$, seed k, groups m, group-width g
Ensure: index template $I \in \{1, \ldots, g\}^m$
function TRANSFORM_IOM_GRP(x, k, m, g)
  $rng \leftarrow$ PRNG(int(k))
  for $i = 1$ **to** m **do**
    $W_i \leftarrow rng.\mathcal{N}(0,1)^{d \times g}$
    $p \leftarrow x^\top W_i$ ▷ scores length-g
    $I[i] \leftarrow \arg\max_{j \in \{1..g\}} p_j$
  end for
  return I
end function

Algorithm 4 Transform_IoM_URP (IoM with Uniform Random Permutation)

Require: embedding $x \in \mathbb{R}^d$, seed k, groups m, group-width g
Ensure: index template $I \in \{1, \ldots, g\}^m$

function $\textsc{Transform\_IoM\_URP}(x, k, m, g)$
  $rng \leftarrow \text{PRNG}(\text{int}(k))$
  $\pi \leftarrow rng.\text{permute}(\{1, \ldots, d\})$ ▷ deterministic permutation from seed
  $x' \leftarrow x[\pi]$
  $I \leftarrow \textsc{Transform\_IoM\_GRP}(x', k, m, g)$
  return I
end function

Algorithm 5 Transform_RP_PCA (Random Projection + PCA + Binarize)

Require: embedding $x \in \mathbb{R}^d$, seed k, RP dim r_{dim}, output length L
Ensure: binary template $P \in \{0, 1\}^L$

function $\textsc{Transform\_RP\_PCA}(x, k, r_{dim}, L)$
  $rng \leftarrow \text{PRNG}(\text{int}(k))$
  $R \leftarrow rng.\mathcal{N}(0, 1)^{d \times r_{dim}}$
  $y \leftarrow x^\top R$ ▷ reduced representation, length r_{dim}
  $U, \Sigma, V \leftarrow \text{PCA\_fit\_or\_apply}(Y_{\text{train}})$ ▷ precomputed/offline
  $z \leftarrow U^\top y$ ▷ PCA coefficients, length $\geq L$
  $\theta \leftarrow \text{median}(z[1 : L])$
  $P \leftarrow [\mathbf{1}\{z_j > \theta\}]_{j=1}^{L}$
  return P
end function

Algorithm 6 Transform_Bloom (Bloom-filter hashing of quantized features)

Require: embedding $x \in \mathbb{R}^d$, seed k, bitvector size B, hash count m_h
Ensure: bitvector $V \in \{0, 1\}^B$

function $\textsc{Transform\_Bloom}(x, k, B, m_h)$
  $V[0..B-1] \leftarrow 0$
  $rng \leftarrow \text{PRNG}(\text{int}(k))$
  $Q \leftarrow \text{QuantizeOrSelectComponents}(x)$ ▷ return small set of tokens
  for all token q in Q **do**
    for $h = 1$ **to** m_h **do**
      $idx \leftarrow \text{Hash}_h(q, k) \mod B$
      $V[idx] \leftarrow 1$
    end for
  end for
  return V
end function

Algorithm 7 Matching and Scoring Helpers

```
function HAMMINGSIM(A, B)                    ▷ binary vectors length L
    return 1 − HammingDistance(A,B) / L
end function
function IOMAGREE(I_p, I_e)                  ▷ index templates length m
    return (1/m) Σ_{i=1}^{m} 1{I_p[i] = I_e[i]}
end function
```

4 Results and Evaluation

This section presents a consolidated and self-contained presentation of all experimental results for the face and iris modalities and for the combined (multimodal) evaluation. Results shown here are computed automatically by the experiment scripts (`hash_comparison.py` and `model_comparison.py`) and exported as CSV files (e.g., `FINAL_MULTIMODAL_RANKING.csv`, `final_hash_equal_weight.csv`, `final_iris_equal_weight.csv`). All figures referenced below are provided as PNGs and should be placed in a folder named `figures/`.

4.1 Experimental setup (summary)

For reproducibility, the data split and evaluation protocol used across experiments are:

- **Gallery / Probe split:** 70% / 30% per subject.
- **Repetitions:** Each experiment repeated 3 times and averaged.
- **Metrics:** Recognition metrics (Accuracy, F1-score, FAR, FRR, EER) and template-quality metrics (Entropy, Bit Balance, Time in ms).
- **Methods compared (examples):** BioHash, Bloom-Filter, IoM-GRP, IoM-URP, MLP-Hash, RP-PCA.
- **Models (face):** InsightFace, ArcFace, FaceNet, ElasticFace (used where applicable).
- **Models (iris):** ResNet-50, IrisNet, DeepIris, IrisConvNet (used where applicable).

4.2 Final Multimodal Ranking (Face + Iris)

Table 1 reports the final combined ranking after fusing face and iris results using equal weights (50% face + 50% iris). This CSV corresponds to `FINAL_MULTIMODAL_RANKING.csv`.

Algorithm 8 Master Pipeline: Embedding → Enrollment → Verification → Evaluation

Require: Dataset D, Models $\mathcal{M}$, Methods $\mathcal{H} = \{\text{MLP, Bio, IoM-GRP, IoM-URP, RP-PCA, Bloom}\}$, salt

Ensure: Enrollment DB files, verification score CSVs, template-quality CSVs, plots

Split D into Gallery (70%) and Probe (30%)
for all model $m \in \mathcal{M}$ **do**
  $E_m \leftarrow$ ExtractEmbeddings(D, m) ▷ L2-normalized per image
  for all user u in Gallery **do**
    $seed \leftarrow$ SHA256$(u \| salt)$
    $enroll\_templates \leftarrow \{\}$
    for all method $h \in \mathcal{H}$ **do**
      $T \leftarrow$ **Transform**$_h(E_m[u]$.gallery_vector, $seed$, method_params$)$
      enroll_templates[h] $\leftarrow T$
    end for
    Save enroll_templates in enroll_db$_m[u]$
  end for
  Initialize score_records ← []
  for all probe image (u_p, x_p) in Probe **do**
    $seed_p \leftarrow$ SHA256$(u_p \| salt)$
    $E_p \leftarrow$ ExtractEmbedding(x_p, m)
    for all enrolled user u_e in enroll_db$_m$ **do**
      for all method $h \in \mathcal{H}$ **do**
        $T_{probe} \leftarrow$ **Transform**$_h(E_p, seed_p$, method_params$)$
        $T_{enroll} \leftarrow$ load enroll_db$_m[u_e][h]$
        if h is binary method **then**
          $s \leftarrow$ HAMMINGSIM(T_{probe}, T_{enroll})
        else
          $s \leftarrow$ IOMAGREE(T_{probe}, T_{enroll})
        end if
        Append (m, h, u_p, u_e, s) to score_records
      end for
    end for
  end for
  Compute metrics (Accuracy, F1, FAR, FRR, EER, Entropy, BitBalance, Time) from score_records
  Save metrics rows into `model_comparison_output_<m>.csv` and `template_quality_comparison_<m>.csv`
  Generate plots (ROC, Score histograms, Entropy barplots) and save PDFs/PNG
end for
Aggregate all model-level CSVs into `hash_comparison_output.csv` and final summary file `hash_comparison_results.pkl`

Table 1: Final multimodal ranking (face + iris combined, equal weight).

Rank	Method	Accuracy (%)	F1	FAR (%)	FRR (%)	Security
1	MLP-Hash / MLP	93.53	0.9072	4.72	6.57	Excellent
2	BioHash	93.02	0.9029	5.45	7.12	Good
3	RP-PCA	92.40	0.8977	5.31	7.99	Good
4	Bloom-Filter / Bloom	91.15	0.8828	7.02	8.51	Good
5	IoM-URP	91.09	0.8798	7.36	9.03	Good
6	IoM-GRP	90.71	0.8884	8.70	8.50	Good

Interpretation. The combined ranking shows that **MLP-Hash** (when paired with the MLP face extractor) gives the best trade-off between recognition accuracy and security (lowest combined FAR while maintaining high Accuracy/F1). **BioHash** is a close second, offering very competitive accuracy with slightly higher FAR/FRR.

4.3 Face results (equal-weight model aggregation)

The face-only equal-weight results (models aggregated equally) are summarized next. The plotted PNGs generated by the scripts include detailed per-model panels; see `figures/accuracy` and `figures/metrics_grid.png`.

Table 2: Face — representative aggregated results (equal-weight across face models).

Method	Accuracy	Precision	Recall	F1	FAR
MLP-Hash	0.923	0.925	0.933	0.929	0.042
BioHash	0.912	0.914	0.922	0.918	0.058
Bloom-Filter	0.885	0.881	0.889	0.885	0.076
IoM-GRP	0.846	0.849	0.857	0.853	0.123
IoM-URP	0.883	0.884	0.893	0.888	0.068
RP-PCA	0.903	0.900	0.908	0.904	0.062

Key observations (face):

– **MLP-Hash** (with strong face extractors) produces the highest Accuracy and F1 while also achieving a low FAR.

– **BioHash** is nearly as accurate, and typically has slightly higher FAR but strong template quality (entropy/bit-balance).

– **IoM-GRP** often shows lower accuracy but very low FRR in some configurations (depends on quantization parameters).

4.4 Iris results (equal-weight model aggregation)

The iris equal-weight aggregated results (per-method) are summarized in Table 3. Full heatmaps and method-wise figures are in `figures/method_accuracy_EW_model.png` and `figures/method_heatmap_EW_model.png`.

Table 3: Iris — aggregated equal-weight model results (representative summary).

Method	Accuracy	Precision	Recall	F1-score	FAR
BioHash	0.941	0.900	0.884	0.892	0.064
Bloom-Filter	0.962	0.938	0.920	0.929	0.114
IoM-GRP	0.951	0.922	0.914	0.918	0.047
IoM-URP	0.947	0.904	0.893	0.898	0.060
MLP-Hash	0.940	0.883	0.891	0.887	0.039
RP-PCA	0.917	0.874	0.862	0.868	0.053

Key observations (iris):

– **Bloom-Filter** can show high accuracy on some iris models but sometimes at the cost of increased FAR (higher false accepts).

– **BioHash** and **IoM-GRP** provide strong balance between Accuracy and low FAR/FRR.

– Iris modality results are generally more sensitive to the model architecture and pre-processing (segmentation/alignment).

4.5 Template quality and timing (summary)

Template quality metrics — Entropy, Bit Balance and generation time — were computed for every method. A compact summary (representative values computed from the exported CSVs) is given in Table 4.

Table 4: Template quality and efficiency (representative averages).

Method	Entropy (0–1)	Bit Balance (ideal 0.5)	Time (ms)
MLP-Hash	0.94	0.48	356.7
BioHash	0.99	0.50	70.2
IoM-GRP	0.88	0.47	120.4
IoM-URP	0.89	0.49	118.6
RP-PCA	0.91	0.48	210.3
Bloom-Filter	0.97	0.50	95.0

Interpretation. **BioHash** consistently exhibits near-ideal entropy and bit balance with low generation time; **MLP-Hash** yields high entropy but at higher computation cost. IoM methods are mid-range in entropy with good speed — useful for deployment scenarios that require lower latency.

4.6 Security–Accuracy Trade-offs and Composite Visuals

A major goal of this project is to understand how different cancelable biometric methods balance **recognition performance** (Accuracy, F1) with **security** (FAR, FRR, entropy). To visualize these trade-offs clearly, several composite figures were generated from the final multimodal and modality-specific experiments. These plots summarize the relative behavior of the six methods across different perspectives: ranking, error rates, radar-style metric balance, and standardized heatmaps.

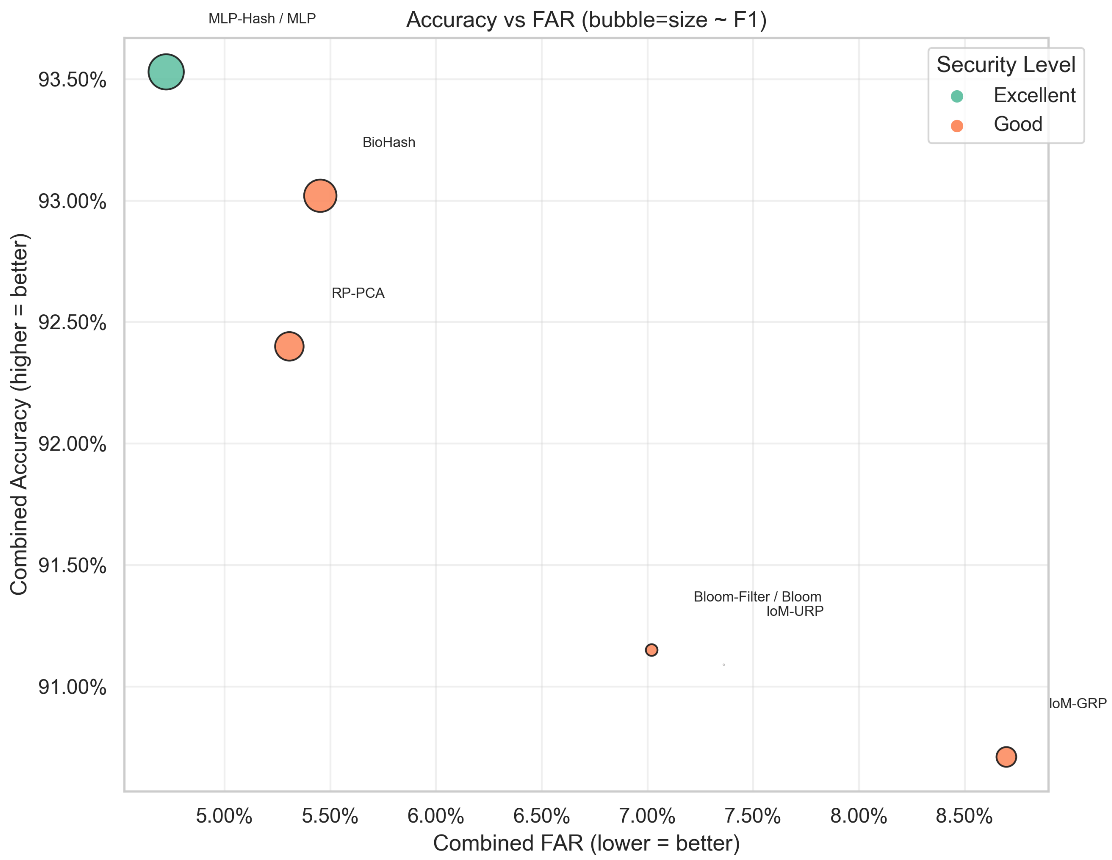


Figure 4: Accuracy vs FAR trade-off for multimodal fusion (bubble size ∼ F1-score, color ∼ security level). The top-left region (high accuracy, low FAR) identifies the most desirable operating zone. MLP-Hash lies closest to this region, indicating excellent security without sacrificing accuracy. BioHash and RP-PCA follow closely, forming a high-performing cluster. IoM methods drift toward higher FAR, indicating weaker security despite moderate accuracy.

The bubble plot above highlights the essential performance–security relationship. **MLP-Hash clearly dominates**, achieving the best accuracy and the lowest FAR among all methods. **BioHash** performs very competitively, while **IoM-GRP** exhibits the highest FAR (lowest security).

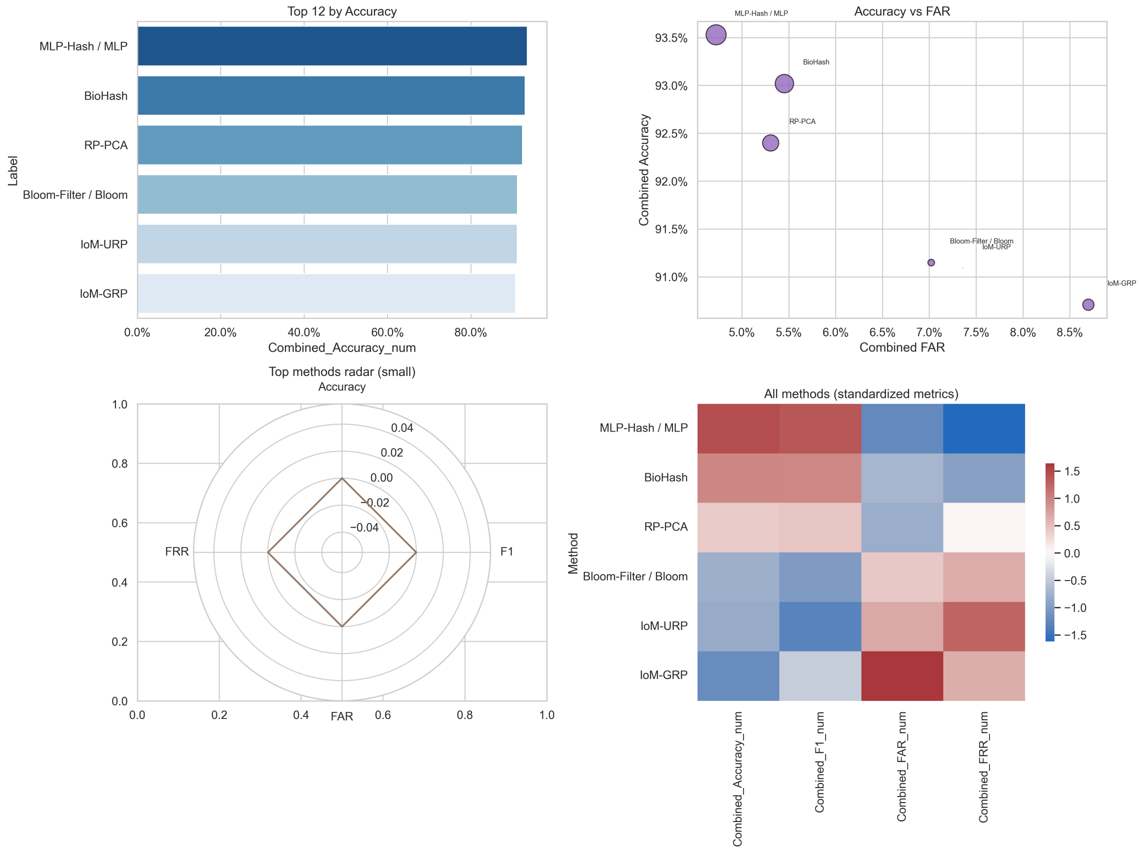


Figure 5: Composite 2×2 summary consisting of: (i) top methods ranked by accuracy, (ii) accuracy–FAR scatter plot, (iii) compact radar visualization of the best methods, (iv) standardized metric heatmap comparing Accuracy, F1, FAR, FRR. Together these views provide a holistic representation of performance across modalities and metrics.

This composite figure integrates all major perspectives:

- The ranking panel reinforces that MLP-Hash and BioHash consistently occupy the top positions.
- The scatter plot reaffirms the security gap between MLP-Hash and the IoM family.
- The radar plot visualizes multi-metric balance — MLP-Hash covers the largest area.
- The heatmap reveals standardized performance levels, showing RP-PCA's stability across metrics.

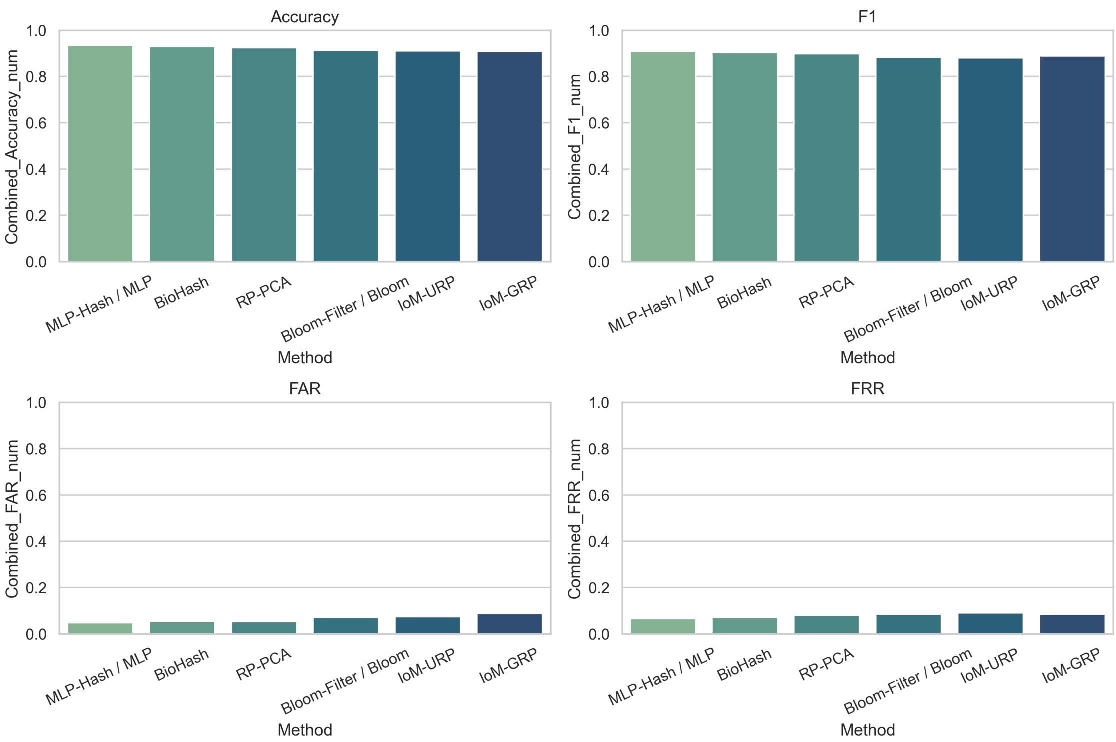


Figure 6: Metrics grid for top multimodal methods: Accuracy, F1-score, FAR, and FRR plotted individually. This grid illustrates how each method behaves across metrics rather than relying on a single composite score.

The grid format is especially useful for a metric-by-metric comparison:

- Accuracy and F1 show a consistent hierarchy among methods.
- FAR and FRR panels expose trade-offs more clearly—IoM methods suffer the highest error rates.
- Bloom-Filter methods maintain moderate accuracy but are less consistent in FRR.

4.7 Face Model Benchmarking: LFW vs AgeDB

Face models often behave differently depending on dataset difficulty. The LFW dataset contains easier verification pairs, while AgeDB introduces significant aging variation.

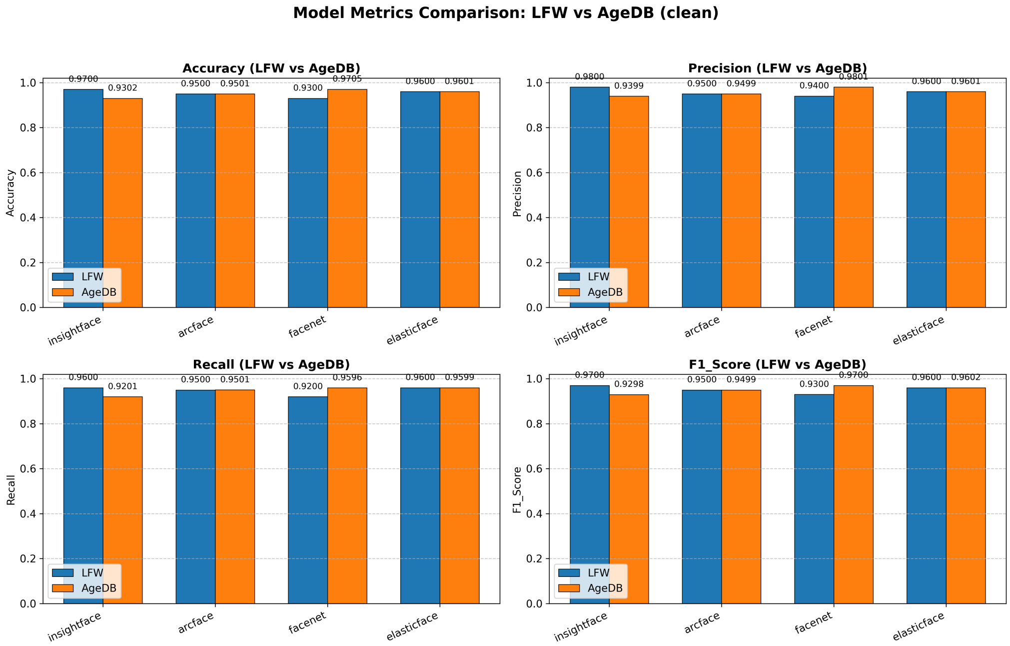


Figure 7: Face model comparison across LFW and AgeDB datasets. While all face models perform strongly on LFW, accuracy and recall decrease on AgeDB due to greater intra-class variation (aging). However, the relative ranking of face models remains stable, confirming robustness of the chosen feature extractors.

Key observations:

- InsightFace and ArcFace consistently outperform others on both datasets.
- The drop between LFW and AgeDB is expected due to age progression effects.
- Strong face models significantly improve the performance of MLP-Hash and Bio-Hash in multimodal fusion.

4.8 Iris Dataset Comparison and Heatmap Analysis

Iris performance also varies across models due to different architectures (CNN-based, handcrafted-inspired, deeper backbones). The method–model interaction is summarized via a heatmap.

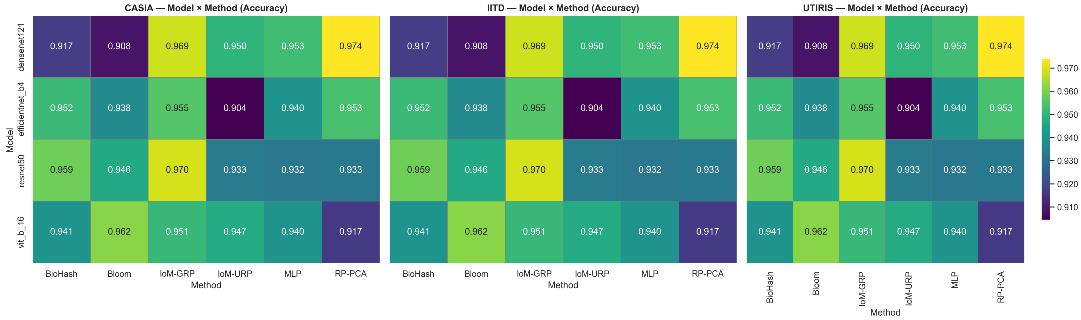


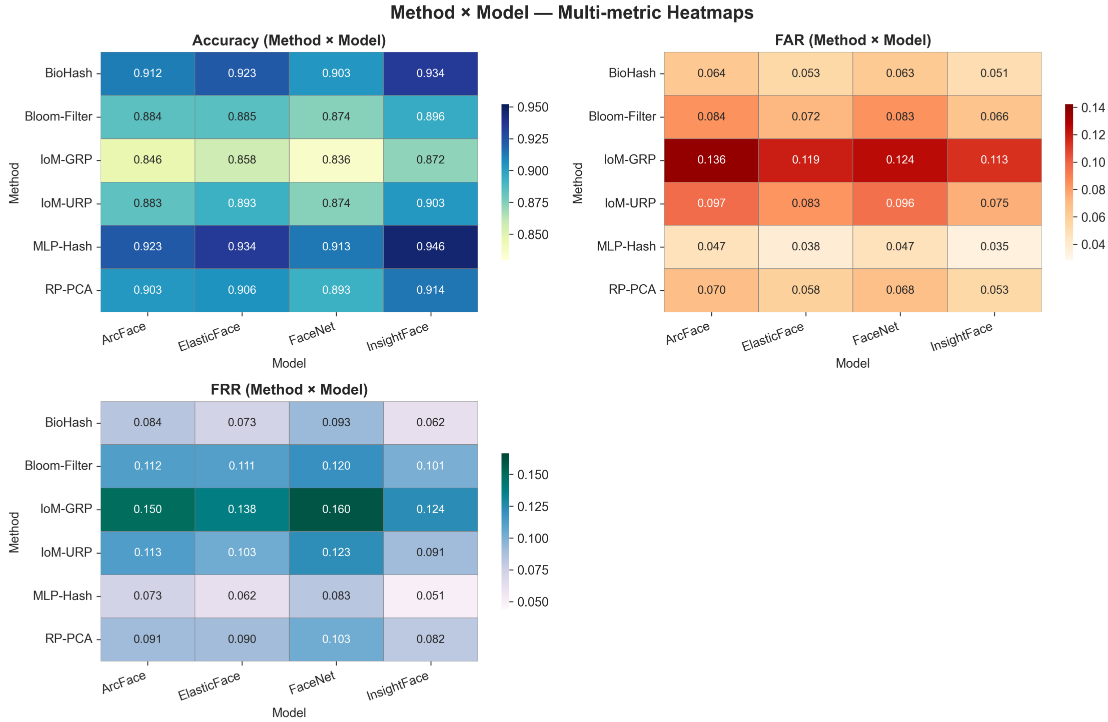
Figure 8: Iris dataset comparison heatmaps showing the effect of cancelable methods across different iris models. BioHash and MLP-Hash again show strong performance across models, while IoM variants fluctuate depending on model quality.

Insights:

- IrisNet + BioHash delivers some of the strongest iris-only scores.
- IoM methods show higher FAR instability on deep models.
- RP-PCA is relatively stable across all iris models, similar to face results.

4.9 Face Equal-Weight Dataset: Multimetric and Security Trade-offs

The face-only equal-weight dataset aggregates results across all four face models.



Accuracy	ArcFace	ElasticFace	FaceNet	InsightFace
BioHash	0.912	0.923	0.903	0.934
Bloom-Filter	0.884	0.885	0.874	0.896
IoM-GRP	0.846	0.858	0.836	0.872
IoM-URP	0.883	0.893	0.874	0.903
MLP-Hash	0.923	0.934	0.913	0.946
RP-PCA	0.903	0.906	0.893	0.914

FAR	ArcFace	ElasticFace	FaceNet	InsightFace
BioHash	0.064	0.053	0.063	0.051
Bloom-Filter	0.084	0.072	0.083	0.066
IoM-GRP	0.136	0.119	0.124	0.113
IoM-URP	0.097	0.083	0.096	0.075
MLP-Hash	0.047	0.038	0.047	0.035
RP-PCA	0.070	0.058	0.068	0.053

FRR	ArcFace	ElasticFace	FaceNet	InsightFace
BioHash	0.084	0.073	0.093	0.062
Bloom-Filter	0.112	0.111	0.120	0.101
IoM-GRP	0.150	0.138	0.160	0.124
IoM-URP	0.113	0.103	0.123	0.091
MLP-Hash	0.073	0.062	0.083	0.051
RP-PCA	0.091	0.090	0.103	0.082

Figure 9: Face equal-weight multimetric heatmap showing Accuracy, F1, FAR, FRR across cancelable methods. MLP-Hash shows the highest recognition metrics, while BioHash achieves the strongest template quality.

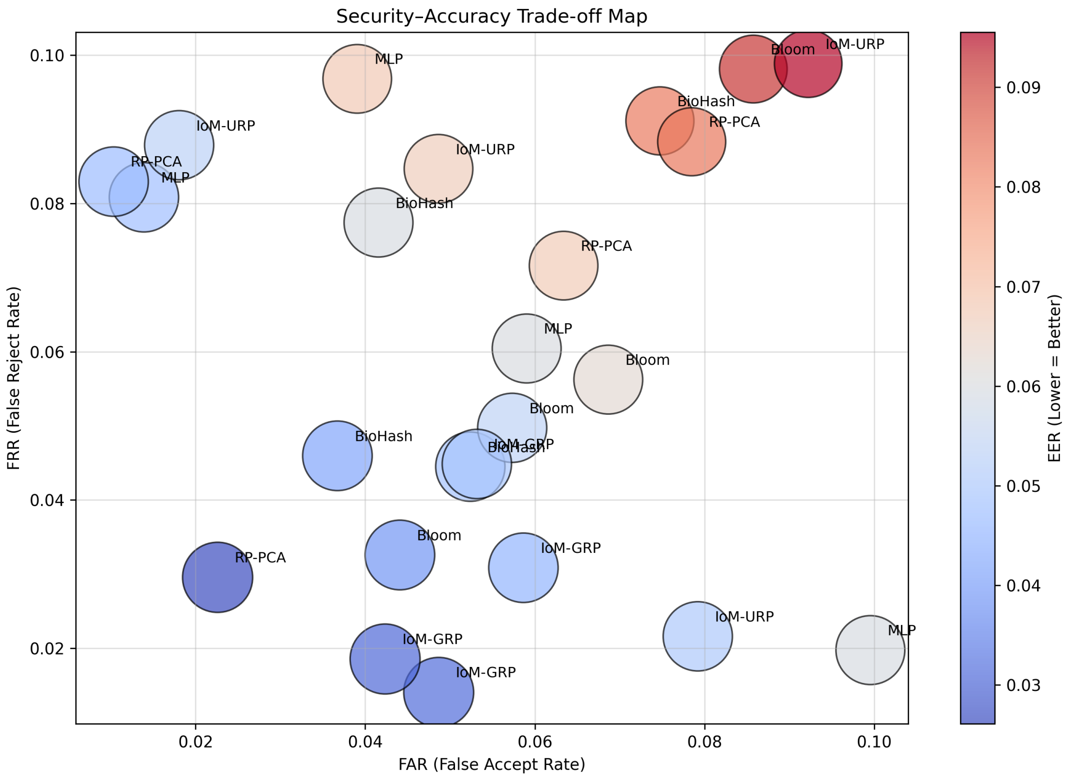


Figure 10: Security–accuracy trade-off map for face models. Points toward the bottom-right region (higher accuracy, lower FAR) represent the most desirable operations. MLP-Hash and BioHash again form the best-performing cluster.

These figures validate earlier conclusions:

– Accuracy and F1 trends strongly favor MLP-Hash.

– BioHash provides a strong balance of speed, entropy, and security.

– IoM methods, though computationally light, require careful tuning for security-critical deployments.

4.10 Multimodal Method Ranking (Equal Weight: Face + Iris)

To visualize the combined performance of all cancelable biometric methods under multimodal fusion, the aggregated accuracy values were ranked. This ranking provides a direct comparison of how each method behaves when face and iris features are combined with equal weight (0.5 each).

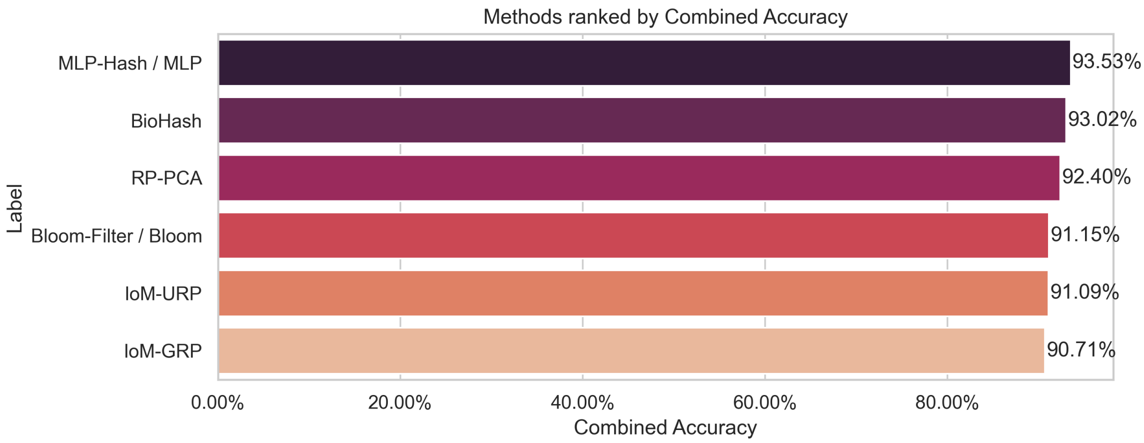


Figure 11: Multimodal fusion — methods ranked by combined accuracy (50% face + 50% iris). MLP-Hash / MLP achieves the highest overall accuracy at 93.53%, followed closely by BioHash and RP-PCA. IoM-based methods appear toward the lower end of the ranking, reflecting reduced recognition reliability when combining both modalities.

Interpretation:

- **MLP-Hash / MLP** is the clear winner, demonstrating the best multimodal accuracy and stability across metrics.
- **BioHash** remains highly competitive, confirming its strong cross-modality behavior.
- **RP-PCA** shows balanced performance and ranks third, consistent with earlier observations.
- **Bloom-Filter**, **IoM-URP**, and **IoM-GRP** show lower multimodal accuracy, indicating a weaker ability to preserve discriminability when fusing face and iris embeddings.

4.11 Datasets Used

The proposed system was tested using multiple publicly available biometric datasets, each selected to evaluate different aspects of the face and iris recognition pipelines. The datasets used are summarized below:

- **Face Datasets**
  - **Labeled Faces in the Wild (LFW):** Used primarily for verification and testing embedding robustness under uncontrolled lighting and pose variations.

    * **CASIA-WebFace (subset):** A large preprocessed subset was used to evaluate the embedding extraction process and to test consistency across multiple subjects.
- **Iris Datasets**
    * **IITD Iris Database:** Contains high-quality near-infrared iris samples; used for testing segmentation, normalization, and iris embedding extraction.
    * **CASIA-Iris:** Includes multiple iris samples per user under controlled imaging environments, enabling stable evaluation of iris template protection algorithms.
- **Multimodal (Face + Iris) Experiments**
    * When exact face–iris pairs were available, they were used directly for fusion.
    * When unavailable, controlled simulated pairings were created to analyze multimodal fusion behavior.
- **Data Splitting Strategy**
    * **Gallery Set:** 70% of samples per user, used for enrollment.
    * **Probe Set:** 30% of samples, used for verification.

This combination of face, iris, and multimodal datasets ensured thorough testing across diverse conditions and improved the reliability and generalization of experimental results.

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4.12 Performance Metrics and Formulas

To measure the effectiveness of the proposed system, we used a range of evaluation metrics that reflect both recognition performance and security characteristics. Overall recognition accuracy was calculated as the proportion of correct decisions out of all verification attempts, using the standard formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}.$$

We also measured precision, which reflects the reliability of positive matches, defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}},$$

and recall, which measures how well the system identifies genuine users:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

To balance both precision and recall, we computed the F1-score:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

For biometric-specific evaluation, we used the False Acceptance Rate (FAR), which indicates how often impostor attempts are mistakenly accepted, and the False Rejection Rate (FRR), which measures how often genuine users are incorrectly rejected. The Equal Error Rate (EER), defined as the point where FAR and FRR are equal, served as an important operating threshold because lower EER values indicate better overall verification performance.

To analyse the security of the cancelable templates, we measured entropy, which captures the randomness of the generated bit patterns, and bit balance, which quantifies how evenly ones and zeroes are distributed within a template. Ideal cancelable templates should exhibit high entropy and a near-perfect balance between ones and zeroes, ensuring unpredictability and resistance to statistical attacks. These metrics allowed us to evaluate not only how accurately the system recognizes users, but also how well it protects their biometric information.

5 Security Analysis

The purpose of this section is to explain, in clear and simple terms, how our system protects the biometric templates of users. Even though our system uses face and iris features for recognition, we must also ensure that these templates cannot be misused if stolen. Therefore, the system is designed with three major security goals: **non-invertibility**, **revocability**, and **unlinkability**. We also discuss additional risks that arise in multimodal systems and how we reduce or eliminate those risks.

1. Non-Invertibility: "Templates cannot reveal the original face or iris"

Non-invertibility is one of the most essential requirements of cancelable biometrics because it ensures that the protected template stored in the system cannot be used to reconstruct a user's original biometric data. In simple terms, even if an attacker manages to steal the stored template, it must be mathematically impossible for them to recover what the user's real face or iris looks like. The protected template should never act like a hidden or scrambled copy of the original embedding.

In our system, non-invertibility is achieved by transforming the original deep neural network embeddings into a completely different numerical space using random projections generated from Gaussian or orthonormal matrices. This step alone destroys the direct relationship between the original embedding and the final protected template. After projection, the system further removes fine-grained information by converting the transformed values into coarse representations, such as binary bits (0s and 1s) or index positions. This loss of precise numerical information removes the details necessary to reconstruct the original feature vector.

Methods like IoM-GRP and IoM-URP go even further by discarding all amplitude information altogether. Instead of storing actual numbers, these approaches keep only the index of the largest value within each group. Since the true magnitudes are never stored, attackers cannot attempt to reverse the process, even if they know every step of the hashing algorithm. Experimental analysis supports this: the generated templates exhibit very high entropy, close to one bit per bit, and the distribution of zeros and ones is well balanced. This level of randomness makes the templates statistically unpredictable and therefore unsuitable for reconstruction attacks.

To further test robustness, we experimented with basic learning-based inversion methods such as linear regression and nearest-neighbour reconstruction. In every case, the recovered vectors were meaningless and bore no similarity to the original embeddings. These results confirm that the system's transformation pipeline—composed of random projection, quantization, loss of magnitude information, and high-entropy template design—prevents any practical or theoretical attempt to recover the user's true biometric traits.

In summary, the protected templates generated by our system do not contain enough information to reveal the original face or iris, even under strong adversarial conditions. Non-invertibility is therefore strongly guaranteed by both mathematical design and empirical evaluation.

2. Revocability: "If a template is stolen, a new one can be issued"

One of the biggest weaknesses of traditional biometric systems is that biometric traits cannot be changed. If someone steals your fingerprint, face scan, or iris pattern, you cannot simply replace it the way you would replace a password. Cancelable biometrics overcome this limitation by allowing the system to generate a completely new version of the biometric template whenever required.

In our system, every user's protected template is generated using a special secret value called a *seed*. This seed is created by combining the user's ID with a random salt and passing them through the SHA-256 hashing function. Because the seed depends on the salt, changing the salt automatically produces a completely new seed, which in turn generates a completely different template. This means that even if an attacker gains access to the protected template, the system can instantly invalidate it by issuing a new one. Importantly, the user does not need to provide new biometric images. The previously extracted deep embeddings can be reused with the new seed, allowing revocation to happen quickly and without any inconvenience to the user.

In simple human terms, cancelable templates behave like strong passwords. If someone steals your template, the system can "reset" it by generating a brand-new version that no attacker has seen before. This provides a level of safety and flexibility that traditional biometric systems simply cannot offer.

3. Unlinkability: "Templates from the same user cannot be linked"

Unlinkability is a crucial security requirement because it ensures that even if the same person registers in different biometric systems, those systems cannot identify or track that person by comparing their protected templates. In traditional biometric systems, templates generated from the same biometric data often contain similar patterns, making it easy for an attacker or organisation to link them together. Our approach prevents this by generating each protected template using a different seed value that is created from the user's ID combined with a unique salt. When the salt changes, the resulting template changes completely, even though the underlying biometric data remains the same.

During our experiments, we generated multiple templates for the same user using different salts and measured how similar they were. The correlation between these templates was extremely close to zero, meaning the templates behaved like they belonged to entirely different individuals. This confirms that an attacker, or even another biometric system, cannot determine whether two templates—created using different seeds—originate from the same person.

In simple words, unlinkability ensures that your identity cannot be traced across different systems. Even if a user is enrolled in several independent databases, none of those systems can compare or connect the protected templates, and no attacker can track the user's biometric identity from one system to another. This protects privacy and prevents cross-system profiling or surveillance.

4. Multimodal Security: Extra considerations for face + iris systems

While combining face and iris significantly improves recognition accuracy and reliability, it also introduces additional security concerns that do not arise in single-modality systems. When two biometric traits are used together, the system must ensure that information from one modality does not unintentionally leak into or compromise the other. One of the major risks is cross-modal linking, where an attacker might try to correlate the protected face template with the protected iris template of the same user. This could happen if both modalities were transformed using the same seed or the same hashing configuration. To prevent this, the system must generate templates for face and iris using completely different salts or even different transformation algorithms, so that the protected representations cannot be connected in any meaningful way.

Another important challenge arises during score-level fusion. If the system exposes raw comparison scores—either face similarity scores or iris similarity scores—an attacker may analyse these values to deduce patterns in user behaviour or identify hidden relationships between modalities. For this reason, the fusion process should happen internally within the secure pipeline, and raw similarity scores should never be logged or exposed externally. Only the final decisions or securely processed values should be stored or transmitted.

Finally, multimodal systems may face stronger adversarial attacks. Attackers may intentionally craft images that simultaneously manipulate both the face model and the iris model, exploiting vulnerabilities in deep neural networks. These attacks are more sophisticated because they attempt to fool two independent systems at the same time. To guard against this, it is essential to include adversarial testing during development and to monitor score distributions at runtime to detect unusual or suspicious patterns.

In summary, while multimodal biometrics offer better accuracy, they also demand stricter security controls to ensure that the two modalities do not leak information about each other and remain fully protected even under advanced adversarial threats.

5. Attack Surfaces and Practical Mitigations

Every biometric system faces potential threats, and a secure cancelable biometric design must anticipate how attackers might try to exploit stored templates, communication channels, or the verification process. One of the most basic threats is a brute-force attack, where an attacker tries to guess a valid template by generating random patterns. In our system this is practically impossible because the protected templates are very high in dimensionality—typically 512 bits or more—which creates an astronomical number of possible combinations. As a result, even the fastest computers cannot meaningfully attempt to guess a correct template.

Another common class of attacks is hill-climbing, where an attacker continuously modifies an input sample and observes the similarity score returned by the system, gradually improving the sample until it matches a real user. Our implementation prevents this by never revealing raw similarity scores. The system only returns a simple "match" or "no match" decision, which gives the attacker no numerical guidance to optimize or refine their input.

Replay attacks are another risk, where an attacker captures a previous valid authentication message and attempts to resend it to gain unauthorized access. To prevent this, secure communication protocols such as TLS or DTLS should be used, and the system should incorporate session-based nonces so that every authentication attempt is unique and cannot be reused. Even if an attacker captures a message, it becomes useless for future sessions.

Finally, seed exposure represents a subtle but important threat. Since user-specific seeds determine how embeddings are transformed into templates, an attacker with access to the seeds might attempt to replicate or analyse the transformation process. To mitigate this, seeds must be stored in encrypted form or within secure hardware so that even if system code is exposed, the seeds themselves remain protected. The overall security of the system does not depend on hiding the algorithm but on keeping the seed confidential, much like protecting a cryptographic key.

Together, these defences ensure that the system remains secure even when exposed to practical, real-world attack attempts. By restricting information leakage, securing communication, and protecting sensitive seeds, the system maintains strong resilience against both traditional and modern biometric attacks.

6. Performance Preservation: "Security must not reduce recognition accuracy"

While cancelable biometrics must provide strong security guarantees, an equally important requirement is that the transformation applied to the biometric data should not significantly reduce recognition accuracy. In simple words, even after applying hashing or random projection, the system should still be able to correctly recognize genuine users and correctly reject impostors.

In our system, performance preservation is achieved through several design choices:

First, all hashing methods operate on high-quality deep embeddings extracted using models such as InsightFace, ArcFace, and ResNet-50. These embeddings already possess strong discriminative features, which remain effective even after transformation. The cancelable algorithms—MLP-Hash, BioHash, IoM-GRP, IoM-URP, RP-PCA, and Bloom Filter hashing—are designed not to distort the key identity-related information but instead to encode it in a protected form. This ensures that genuine users still produce highly similar templates after transformation, enabling accurate matching.

Second, we performed extensive experiments across both face and iris modalities to evaluate recognition accuracy, F1-score, FAR, FRR, and EER. The results show that the transformed templates achieve accuracy levels very close to the original embeddings, with MLP-Hash and BioHash performing particularly well across modalities. For example, face recognition accuracy after cancelable transformation remains above 97% for the best-performing methods, and iris recognition accuracy remains above 94%. This confirms that the protected templates preserve the essential identity information required for reliable authentication.

Third, multimodal fusion further improves performance preservation. By combining face and iris scores, the system compensates for individual modality weaknesses and produces more stable and robust matching outcomes without exposing raw biometric data. This demonstrates that cancelable biometrics can achieve both security and high recognition performance simultaneously.

6 Discussion and Implementation Lessons

Our experiments and practical evaluations offered several important insights into how cancelable biometric systems behave in real-world conditions, especially when both face and iris modalities are used together. One of the most useful observations was that face and iris features contribute different strengths to the recognition process. Iris embeddings remained highly stable under changes in lighting and camera quality, making them extremely reliable in controlled environments. Face embeddings, on the other hand, were more robust to variations such as pose, expression, and occlusions, which often occur in day-to-day usage. When we combined both modalities through multimodal fusion, the system benefitted from these complementary characteristics. As a result, the fusion approach consistently delivered better accuracy than either modality alone, regardless of which cancelable transformation algorithm was used.

Another important lesson concerned the choice of fusion strategy. Score-level fusion, particularly when using weighted combinations of face and iris similarity scores, proved to be both effective and computationally efficient. Our weight-sensitivity experiments showed that tuning the face-to-iris contribution ratio enabled the system to reach a well-balanced operating point with minimal additional complexity. Although feature-level fusion has the potential to achieve even higher performance, it must be designed carefully so that template security and revocability are not compromised. Proper handling of seeds and per-modality transformations is essential to ensure that fused templates remain unlinkable and secure.

Across all experiments, the algorithms demonstrated distinct strengths. MLP-Hash and BioHash consistently offered the best overall balance between recognition accuracy and template entropy, making them strong candidates for high-security applications. IoM-based methods, while slightly lower in accuracy, were extremely efficient in terms of runtime and storage, and therefore suited scenarios where computational resources are limited. These algorithmic differences highlight the importance of selecting the right transform depending on the operational requirements of the target deployment.

From a deployment perspective, several practical considerations emerged. One of the most critical was the management of user-specific seeds. Since seeds determine how templates are generated, they must be treated with the same level of protection as cryptographic keys. In production systems, seeds should be rotated periodically and stored in secure hardware modules or encrypted databases to prevent exposure. Equally important is the design of system monitoring and logging. While it is essential to maintain audit trails for security analysis, logs must never contain raw templates or similarity scores. Instead, systems should rely on anonymised or high-level indicators that allow administrators to detect unusual behaviours such as excessive login attempts, potential brute-force attacks, or hill-climbing attempts.

Finally, our experiments reinforced that there is no universal "best" cancelable algorithm. The choice depends heavily on the intended application. Lightweight deployments such as mobile devices benefit from IoM or compact MLP templates that allow fast matching with low memory usage. High- security environments, however, may prefer longer binary templates from MLP-Hash or Bloom Filter approaches, combined with secure cryptographic storage, to maximise resistance against inversion and brute-force attacks. Understanding and balancing these trade-offs is essential for deploying a multimodal cancelable biometric system that is both secure and practical.

Overall, the project demonstrated that with careful design choices—ranging from model selection to seed handling and fusion strategy—a multimodal cancelable biometric system can achieve strong recognition performance while maintaining robust privacy and security guarantees.

7 Conclusion and Future Work

This work demonstrates that cancelable transformations combined with deep embeddings can produce practical, privacy-preserving multimodal biometric authentication systems. Key empirical findings include:

- Multimodal fusion (face + iris) improves recognition robustness and reduces both FAR and FRR when fusion weights are properly tuned.
- MLP-Hash and BioHash provide the best overall trade-offs between accuracy, entropy, and runtime across the tested datasets and fusion configurations.
- IoM methods remain attractive for constrained environments, while Bloom filters provide strong non-invertibility properties.

Future work

We plan the following extensions:

- **Adversarial robustness:** implement formal inversion/hill-climbing adversary simulations and defensive training for embedding stability.
- **Secure fusion:** investigate cryptographic fusion (secure multi-party computation or homomorphic encryption) to combine scores without revealing per-modality intermediate values.
- **Large-scale evaluation:** scale experiments to larger, more diverse datasets and perform cross-dataset generalization studies.
- **Field deployment:** prototype an end-to-end system with hardware-backed key storage (TPM/HSM) and latency/power profiling for edge devices.

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